Reg. No										

St. Joseph's College of Arts & Science (Autonomous)
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MT408 – GRAPH THEORY

Time : 3 hrs **Max Marks :75**

SECTION – A (5X2=10)

Answer **ALL** Questions

1. Define complete graph.
2. Define product of two graphs G_1 and G_2.
3. Prove that there is no 3-connected graph with 7 edges.
4. Prove that every Hamiltonian graph is 2-connected.
5. Define chromatic number.

SECTION – B (3X5=15)

Answer any **THREE** Questions

6. Prove that any self complementary graphs has 4n or 4n+1 points.
7. Prove that in a graph G, any u – v walk contains a u – v path.
8. Prove that every non-trivial connected graphs has at least two points which are not cutpoints.
9. State and prove necessary condition for a graph G to be Hamiltonian graph.
10. If G is uniquely n–colourable, then prove that $\delta(G) > n-1$.

SECTION – C (5X10=50)

Answer **ALL** Questions

11. a) i) Show that in any group of two or more people ,there are atleast two with exactly the same number of friends inside] the group.

ii) Prove that in any graph G the number of points of odd degree is even.

(or)

b) i) Prove that every graph is an intersection graph.

ii) Let G be a (p , q) graph , then prove that L (G) is a (q , q_L) graph where $qL = \frac{1}{2}\left(\sum_{i=1}^{p} d_i^2\right) - q.$

12. a) If a partition P = (d_1 , d_2 ,……, d_P) with $d_1 \geq d_2 \geq ..... \geq d_p$ is graphical , then prove that $\sum_{i=1}^{p} d_i$ is even and $\sum_{i=1}^{k} d_i \leq k$ (k-1) + $\sum_{i=k+1}^{p}$ min { k , d_i } for $1 \leq k \leq p$.

(or)

b) i) If $\delta \geq k$, then prove that G has a path of length k .

ii) Prove that a closed walk of odd length contains a cycle.

13. a) Prove that a graph G is connected if and only if for any partition of V into subsets V_1 and V_2 there is a line of G joining a point of V_1 to a point of V_2 .

(or)

b) Let x be a line of a connected graph G, then prove that the following statements are equivalent .

i) x is a bridge of G .

ii) There exists a partition of V in to two subsets U and W such that for every point $u \in U$ and $w \in W$,the line x is on every u – w path.

iii) There exist two points u, w such that the line x is on every u – w path.

14. a) If G is a graph with $p \geq 3$ vertices and $\delta \geq \frac{p}{2}$, then prove that G is Hamiltonian.

(or)

b) Let G be a (p, q) graph, then prove that the following statements are equivalent.
i) G is a tree.
ii) Every two points of G are joined by a unique path.
iii) G is connected and p = q + 1.
iv) G is acyclic and p = q +1.

15. a) Prove that a graph G can be embedded in the surface of a sphere if and only if it can be embedded in a plane.

(or)

b) Prove that the following statements are equivalent for any graph G.
i) G is 2 –colourable.
ii) G is bipartite.
ii) Every cycle of G has even length.

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